

• 卡尔曼滤波公式与计算步骤

先验估计

$$\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1} + Bu_k$$

先验误差矩阵

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q$$

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$$K_k = P_{k|k-1}H^T(HP_{k|k-1}H^T + R)^{-1}$$

后验估计

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - H\hat{x}_{k|k-1})$$

更新协方差矩阵

$$P_{k|k} = (I - K_kH)P_{k|k-1}$$

其中 $A, B, H, Q, R, \hat{x}_{0|0}, P_{0|0}$ 需自行给定。

• 扩展卡尔曼滤波

扩展卡尔曼滤波在形式上与卡尔曼滤波类似,但是它事实上不能直接从卡尔曼滤波的数学公式推出来,我们这里不去推导理论公式,只说明如何将它运用在无感 FOC 控制中

卡尔曼滤波使用线性方程(下左),而扩展卡尔曼滤波使用非线性方程(下右):

$$\begin{cases} x_k = Ax_{k-1} + Bu_k + w_k \\ z_k = Hx_k + v_k \end{cases}, \begin{cases} x_k = f(x_{k-1}, u_k) + w_k \\ z_k = h(x_k) + v_k \end{cases}$$

由于我们的 u_k 为线性的,我们也把上述式子写成:

$$\begin{cases} x_k = f(x_{k-1}) + Bu_k \\ y_k = h(x_k) \end{cases}$$

其中 w_k 和 v_k 为过程噪声和测量噪声,仅在卡尔曼滤波推导过程中使用,我们只是使用该公式,不需要带上这两项。另外,输出变量在理论推导时我们写成 z_k ,使用时一般写成 y_k

永磁同步电机的静止坐标系中的定子电压方程为:

$$\begin{cases} u_\alpha = Ri_\alpha + \frac{d\psi_\alpha}{dt} = Ri_\alpha + L_s \frac{di_\alpha}{dt} - \omega_e \psi_f \sin\theta_e \\ u_\beta = Ri_\beta + \frac{d\psi_\beta}{dt} = Ri_\beta + L_s \frac{di_\beta}{dt} + \omega_e \psi_f \cos\theta_e \end{cases}$$

上述方程可以变换成:

$$\begin{cases} \frac{di_\alpha}{dt} = \frac{u_\alpha}{L_s} - \frac{R}{L_s} i_\alpha + \frac{1}{L_s} \omega_e \psi_f \sin\theta_e \\ \frac{di_\beta}{dt} = \frac{u_\beta}{L_s} - \frac{R}{L_s} i_\beta - \frac{1}{L_s} \omega_e \psi_f \cos\theta_e \\ \frac{d\omega_e}{dt} = 0 \\ \frac{d\theta_e}{dt} = \omega_e \end{cases}$$

其中 $\frac{d\omega_e}{dt} = 0$ 和 $\frac{d\theta_e}{dt} = \omega_e$ 是我们自己加上去的,我们使用了简化的模型,认为 $\frac{d\omega_e}{dt} = 0$ 即 $\frac{d\omega_e}{dt}$ 变化缓慢。

输入变量,状态变量,输出变量我们取:

$$u = \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix}, x = \begin{bmatrix} i_\alpha \\ i_\beta \\ \omega_e \\ \theta_e \end{bmatrix}, y = \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}$$

得到：

$$\dot{x}_k = g(x_{k-1}) + \begin{bmatrix} \frac{1}{L_s} & 0 \\ 0 & \frac{1}{L_s} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} u_k, \text{其中 } g(x_k) = \begin{bmatrix} -\frac{R}{L_s} i_\alpha + \frac{1}{L_s} \omega_e \psi_f \sin \theta_e \\ -\frac{R}{L_s} i_\beta - \frac{1}{L_s} \omega_e \psi_f \cos \theta_e \\ 0 \\ \omega_e \end{bmatrix}$$

使用后向欧拉离散化（EKF 执行周期设为 T_s ）：

$$\frac{x_k - x_{k-1}}{T_s} \approx \dot{x}_k = g(x_{k-1}) + \begin{bmatrix} \frac{1}{L_s} & 0 & 0 & 0 \\ 0 & \frac{1}{L_s} & 0 & 0 \end{bmatrix}^T u_k$$

$$x_k = x_{k-1} + T_s \left[g(x_{k-1}) + \begin{bmatrix} \frac{1}{L_s} & 0 & 0 & 0 \\ 0 & \frac{1}{L_s} & 0 & 0 \end{bmatrix}^T u_k \right]$$

把上述离散方程写成以下形式：

$$x_k = f(x_{k-1}) + Bu_k, y_k = Cx_k$$

其中：

$$f(x) = x + T_s g(x), B = T_s \begin{bmatrix} \frac{1}{L_s} & 0 & 0 & 0 \\ 0 & \frac{1}{L_s} & 0 & 0 \end{bmatrix}^T, H = C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

我们本来拥有 $\hat{x}_{k-1|k-1}$ ，可以直接算出先验估计

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}) + Bu_k$$

但是在计算先验误差矩阵时

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q$$

我们缺少矩阵 **A**，扩展卡尔曼滤波的做法是使用雅克比矩阵代替 **A**：

$$\begin{aligned}
A &= \frac{\partial f}{\partial x}|_{(\hat{x}_{k-1|k-1}, u_k, 0)} \\
&= \begin{bmatrix} 1 - T_s \frac{R}{L_s} & 0 & T_s \frac{1}{L_s} \psi_f \sin \theta_e & T_s \frac{1}{L_s} \omega_e \psi_f \cos \theta_e \\ 0 & 1 - T_s \frac{R}{L_s} & -T_s \frac{1}{L_s} \psi_f \cos \theta_e & T_s \frac{1}{L_s} \omega_e \psi_f \sin \theta_e \\ 0 & 0 & 1 & 0 \\ 0 & 0 & T_s & 1 \end{bmatrix}|_{(\hat{x}_{k-1|k-1}, u_k, 0)} \\
&= \begin{bmatrix} 1 - T_s \frac{R}{L_s} & 0 & T_s \frac{1}{L_s} \psi_f \sin \theta_{e(k-1)} & T_s \frac{1}{L_s} \omega_{e(k-1)} \psi_f \cos \theta_{e(k-1)} \\ 0 & 1 - T_s \frac{R}{L_s} & -T_s \frac{1}{L_s} \psi_f \cos \theta_{e(k-1)} & T_s \frac{1}{L_s} \omega_{e(k-1)} \psi_f \sin \theta_{e(k-1)} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & T_s & 1 \end{bmatrix}
\end{aligned}$$

此时可计算先验误差矩阵

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q$$

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$$K_k = P_{k|k-1}H^T(HP_{k|k-1}H^T + R)^{-1} = P_{k|k-1}C^T(CP_{k|k-1}C^T + R)^{-1}$$

后验估计

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - h(\tilde{x}_{k-1}, 0)) = \hat{x}_{k|k-1} + K_k(y_k - C\tilde{x}_{k-1})$$

其中

$$\tilde{x}_{k-1} = f(\hat{x}_{k-1|k-1}, u_k, 0) = \hat{x}_{k|k-1}$$

更新协方差矩阵

$$P_{k|k} = (I - K_kC)P_{k|k-1}$$

其中

$$y_k = z_k, B = T_s \begin{bmatrix} \frac{1}{L_s} & 0 \\ 0 & \frac{1}{L_s} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, H = C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

• 矩阵展开和优化

由于直接使用 for for for 来计算矩阵非常的耗时，一次运算需要接近 200us，满足不了 20khz 的 FOC（50us 周期），因此需要手动展开矩阵

遵循两个原则：实对称矩阵只存一半的变量，矩阵中为 0 的变量不存储，为常数的变量选择性地存储，常数的乘除法可以合并的就合并，提前计算好

先验估计

$$\begin{aligned}
\hat{x}_{k|k-1} &= f(\hat{x}_{k-1|k-1}) + Bu_k \\
\hat{x}_{k|k-1} &= \begin{bmatrix} i_{\alpha(k-1)} \\ i_{\beta(k-1)} \\ \omega_{e(k-1)} \\ \theta_{e(k-1)} \end{bmatrix} + T_s \begin{bmatrix} -\frac{R}{L_s} i_{\alpha(k-1)} + \frac{1}{L_s} \omega_{e(k-1)} \psi_f \sin \theta_{e(k-1)} + \frac{1}{L_s} u_{k-1}[0] \\ -\frac{R}{L_s} i_{\beta(k-1)} - \frac{1}{L_s} \omega_{e(k-1)} \psi_f \cos \theta_{e(k-1)} + \frac{1}{L_s} u_{k-1}[1] \\ 0 \\ \omega_{e(k-1)} \end{bmatrix}
\end{aligned}$$

```

const float i_alpha_k_prev = x_k_k[0];
const float i_beta_k_prev = x_k_k[1];
const float w_e_k_prev = x_k_k[2];
const float theta_e_k_prev = x_k_k[3];

float c_theta_e_k_prev, s_theta_e_k_prev; // 不能为 const

static const float RAD_TO_DEG = 180.0f / PI;
arm_sin_cos_f32(theta_e_k_prev * RAD_TO_DEG, &s_theta_e_k_prev,
&c_theta_e_k_prev);
const float w_flux = w_e_k_prev * flux;
x_k_k_prev[0] = i_alpha_k_prev +
                (-Rs * i_alpha_k_prev + w_flux * s_theta_e_k_prev +
ekf->Valpha_I)*TsDivLs;
x_k_k_prev[1] = i_beta_k_prev +
                (-Rs * i_beta_k_prev - w_flux * c_theta_e_k_prev +
ekf->Vbeta_I)*TsDivLs;
x_k_k_prev[2] = w_e_k_prev;
x_k_k_prev[3] = theta_e_k_prev + T_s * w_e_k_prev;

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Ak 矩阵

$$A_k = \frac{\partial f}{\partial x|_{(\hat{x}_{k-1|k-1}, u_k, 0)}}$$

$$= \begin{bmatrix} 1 - T_s \frac{R}{L_s} & 0 & T_s \frac{1}{L_s} \psi_f \sin \theta_{e(k-1)} & T_s \frac{1}{L_s} \omega_{e(k-1)} \psi_f \cos \theta_{e(k-1)} \\ 0 & 1 - T_s \frac{R}{L_s} & -T_s \frac{1}{L_s} \psi_f \cos \theta_{e(k-1)} & T_s \frac{1}{L_s} \omega_{e(k-1)} \psi_f \sin \theta_{e(k-1)} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & T_s & 1 \end{bmatrix}$$

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const float A_k_0_0=1 - Rs*T_sDivLs;
const float A_k_1_1=1 - Rs*T_sDivLs;
const float A_k_3_2=T_s;
const float TsDivLsFlux=TsDivLs*flux;
const float A_k_0_2= TsDivLsFlux*s_theta_e_k_prev;
const float A_k_1_2= - TsDivLsFlux * c_theta_e_k_prev;
const float A_k_0_3= -A_k_1_2*w_e_k_prev;
const float A_k_1_3 = A_k_0_2*w_e_k_prev;

```

先验方差矩阵

$$P_{k|k-1} = A_k P_{k-1|k-1} A_k^T + W_k Q W_k^T = A_k P_{k-1|k-1} A_k^T + Q$$

这里设 Q 为对角矩阵，然后设定 $P_{0|0}$ 为实对称阵，因此 $P_{k|k-1}$ 为实对称阵。

$$P_{k|k-1} = \begin{bmatrix} a_{k(0,0)} & 0 & a_{k(0,2)} & a_{k(0,3)} \\ 0 & a_{k(1,1)} & a_{k(1,2)} & a_{k(1,3)} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & a_{k(3,2)} & 1 \end{bmatrix} \\ * \begin{bmatrix} P_{k-1|k-1(0,0)} & P_{k-1|k-1(0,1)} & P_{k-1|k-1(0,2)} & P_{k-1|k-1(0,3)} \\ P_{k-1|k-1(1,0)} & P_{k-1|k-1(1,1)} & P_{k-1|k-1(1,2)} & P_{k-1|k-1(1,3)} \\ P_{k-1|k-1(2,0)} & P_{k-1|k-1(2,1)} & P_{k-1|k-1(2,2)} & P_{k-1|k-1(2,3)} \\ P_{k-1|k-1(3,0)} & P_{k-1|k-1(3,1)} & P_{k-1|k-1(3,2)} & P_{k-1|k-1(3,3)} \end{bmatrix} A_k^T + Q$$

使用分块矩阵计算 $A_k P_{k-1|k-1} A_k^T$:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} P_A & P_B \\ P_C & P_D \end{bmatrix} \begin{bmatrix} A^T & C^T \\ B^T & D^T \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D \end{bmatrix} \begin{bmatrix} P_A & P_B \\ P_C & P_D \end{bmatrix} \begin{bmatrix} A & 0 \\ B^T & D^T \end{bmatrix} \\ = \begin{bmatrix} AP_A + BP_C & AP_B + BP_D \\ DP_C & DP_D \end{bmatrix} \begin{bmatrix} A & 0 \\ B^T & D^T \end{bmatrix} \\ = \begin{bmatrix} AP_A A + BP_C A + AP_B B^T + BP_D B^T & AP_B D^T + BP_D D^T \\ DP_C A + DP_D B^T & DP_D D^T \end{bmatrix} \\ = \begin{bmatrix} AP_A A + BP_B^T A + AP_B B^T + BP_D B^T & AP_B D^T + BP_D D^T \\ DP_C A + DP_D B^T & DP_D D^T \end{bmatrix}$$

由于上述计算结果仍为对称矩阵，因此我们只需要计算右上角的值。

$$\begin{aligned} AP_A A + BP_B^T A + AP_B B^T + BP_D B^T &= AP_A A + BP_B^T A + (BP_B^T A)^T + BP_D B^T \\ AP_A A &= \begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix} \begin{bmatrix} P_{k-1|k-1(0,0)} & P_{k-1|k-1(0,1)} \\ P_{k-1|k-1(1,0)} & P_{k-1|k-1(1,1)} \end{bmatrix} \begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix} \\ &= \begin{bmatrix} a_{k(0,0)} P_{k-1|k-1(0,0)} & a_{k(0,0)} P_{k-1|k-1(0,1)} \\ a_{k(1,1)} P_{k-1|k-1(1,0)} & a_{k(1,1)} P_{k-1|k-1(1,1)} \end{bmatrix} \begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix} \\ &= \begin{bmatrix} a_{k(0,0)} a_{k(0,0)} P_{k-1|k-1(0,0)} & a_{k(0,0)} a_{k(1,1)} P_{k-1|k-1(0,1)} \\ a_{k(0,0)} a_{k(1,1)} P_{k-1|k-1(1,0)} & a_{k(1,1)} a_{k(1,1)} P_{k-1|k-1(1,1)} \end{bmatrix} \\ BP_D B^T &= \begin{bmatrix} a_{k(0,2)} & a_{k(0,3)} \\ a_{k(1,2)} & a_{k(1,3)} \end{bmatrix} \begin{bmatrix} P_{k-1|k-1(2,2)} & P_{k-1|k-1(2,3)} \\ P_{k-1|k-1(3,2)} & P_{k-1|k-1(3,3)} \end{bmatrix} \begin{bmatrix} a_{k(0,2)} & a_{k(1,2)} \\ a_{k(0,3)} & a_{k(1,3)} \end{bmatrix} \\ &= \begin{bmatrix} a_{k(0,2)} P_{k-1|k-1(2,2)} + a_{k(0,3)} P_{k-1|k-1(2,3)} & a_{k(0,2)} P_{k-1|k-1(2,3)} + a_{k(0,3)} P_{k-1|k-1(3,3)} \\ a_{k(1,2)} P_{k-1|k-1(2,2)} + a_{k(1,3)} P_{k-1|k-1(2,3)} & a_{k(1,2)} P_{k-1|k-1(2,3)} + a_{k(1,3)} P_{k-1|k-1(3,3)} \end{bmatrix} \begin{bmatrix} a_{k(0,2)} & a_{k(1,2)} \\ a_{k(0,3)} & a_{k(1,3)} \end{bmatrix} \\ &= \begin{bmatrix} BPB_{(0,0)} & BPB_{(0,1)} \\ BPB_{(1,0)} & BPB_{(1,1)} \end{bmatrix} \\ BPB_{(0,0)} &= a_{k(0,2)} a_{k(0,2)} P_{k-1|k-1(2,2)} + 2a_{k(0,2)} a_{k(0,3)} P_{k-1|k-1(2,3)} + a_{k(0,3)} a_{k(0,3)} P_{k-1|k-1(3,3)} \\ BPB_{(0,1)} &= BPB_{(1,0)} \\ &= a_{k(1,2)} (a_{k(0,2)} P_{k-1|k-1(2,2)} + a_{k(0,3)} P_{k-1|k-1(2,3)}) + a_{k(1,3)} (a_{k(0,2)} P_{k-1|k-1(2,3)} \\ &\quad + a_{k(0,3)} P_{k-1|k-1(3,3)}) \\ BPB_{(1,1)} &= a_{k(1,2)} a_{k(1,2)} P_{k-1|k-1(2,2)} + 2a_{k(1,2)} a_{k(1,3)} P_{k-1|k-1(2,3)} + a_{k(1,3)} a_{k(1,3)} P_{k-1|k-1(3,3)} \\ BP_B^T A &= \begin{bmatrix} a_{k(0,2)} & a_{k(0,3)} \\ a_{k(1,2)} & a_{k(1,3)} \end{bmatrix} \begin{bmatrix} P_{k-1|k-1(0,2)} & P_{k-1|k-1(1,2)} \\ P_{k-1|k-1(0,3)} & P_{k-1|k-1(1,3)} \end{bmatrix} \begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix} \\ &= \begin{bmatrix} a_{k(0,2)} P_{k-1|k-1(0,2)} + a_{k(0,3)} P_{k-1|k-1(0,3)} & a_{k(0,2)} P_{k-1|k-1(1,2)} + a_{k(0,3)} P_{k-1|k-1(1,3)} \\ a_{k(1,2)} P_{k-1|k-1(0,2)} + a_{k(1,3)} P_{k-1|k-1(0,3)} & a_{k(1,2)} P_{k-1|k-1(1,2)} + a_{k(1,3)} P_{k-1|k-1(1,3)} \end{bmatrix} \begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix} \\ &= \begin{bmatrix} a_{k(0,0)} (a_{k(0,2)} P_{k-1|k-1(0,2)} + a_{k(0,3)} P_{k-1|k-1(0,3)}) & a_{k(1,1)} (a_{k(0,2)} P_{k-1|k-1(1,2)} + a_{k(0,3)} P_{k-1|k-1(1,3)}) \\ a_{k(0,0)} (a_{k(1,2)} P_{k-1|k-1(0,2)} + a_{k(1,3)} P_{k-1|k-1(0,3)}) & a_{k(1,1)} (a_{k(1,2)} P_{k-1|k-1(1,2)} + a_{k(1,3)} P_{k-1|k-1(1,3)}) \end{bmatrix} \end{aligned}$$

因此结果为:

$$\begin{aligned}
AP_A A + BP_B^T A + AP_B B^T + BP_D B^T &= \begin{bmatrix} P_{k|k-1(0,0)} & P_{k-1|k-1(0,1)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} \end{bmatrix} \\
P_{k|k-1(0,0)} &= a_{k(0,0)} a_{k(0,0)} P_{k-1|k-1(0,0)} + 2a_{k(0,0)} (a_{k(0,2)} P_{k-1|k-1(0,2)} + a_{k(0,3)} P_{k-1|k-1(0,3)}) \\
&\quad + a_{k(0,2)} a_{k(0,2)} P_{k-1|k-1(2,2)} + 2a_{k(0,2)} a_{k(0,3)} P_{k-1|k-1(2,3)} \\
&\quad + a_{k(0,3)} a_{k(0,3)} P_{k-1|k-1(3,3)} \\
P_{k|k-1(0,1)} &= a_{k(0,0)} a_{k(1,1)} P_{k-1|k-1(0,1)} + a_{k(1,1)} (a_{k(0,2)} P_{k-1|k-1(1,2)} + a_{k(0,3)} P_{k-1|k-1(1,3)}) \\
&\quad + a_{k(0,0)} (a_{k(1,2)} P_{k-1|k-1(0,2)} + a_{k(1,3)} P_{k-1|k-1(0,3)}) + a_{k(1,2)} (a_{k(0,2)} P_{k-1|k-1(2,2)} \\
&\quad + a_{k(0,3)} P_{k-1|k-1(2,3)}) + a_{k(1,3)} (a_{k(0,2)} P_{k-1|k-1(2,3)} + a_{k(0,3)} P_{k-1|k-1(3,3)}) \\
P_{k|k-1(1,1)} &= a_{k(1,1)} a_{k(1,1)} P_{k-1|k-1(1,1)} + 2a_{k(1,1)} (a_{k(1,2)} P_{k-1|k-1(1,2)} + a_{k(1,3)} P_{k-1|k-1(1,3)}) \\
&\quad + a_{k(1,2)} a_{k(1,2)} P_{k-1|k-1(2,2)} + 2a_{k(1,2)} a_{k(1,3)} P_{k-1|k-1(2,3)} \\
&\quad + a_{k(1,3)} a_{k(1,3)} P_{k-1|k-1(3,3)}
\end{aligned}$$

$$(AP_B + BP_D)D^T$$

$$\begin{aligned}
AP_B + BP_D &= \begin{bmatrix} a_{k(0,0)} & 0 \\ 0 & a_{k(1,1)} \end{bmatrix} \begin{bmatrix} P_{k-1|k-1(0,2)} & P_{k-1|k-1(0,3)} \\ P_{k-1|k-1(1,2)} & P_{k-1|k-1(1,3)} \end{bmatrix} \\
&\quad + \begin{bmatrix} a_{k(0,2)} & a_{k(0,3)} \\ a_{k(1,2)} & a_{k(1,3)} \end{bmatrix} \begin{bmatrix} P_{k-1|k-1(2,2)} & P_{k-1|k-1(2,3)} \\ P_{k-1|k-1(3,2)} & P_{k-1|k-1(3,3)} \end{bmatrix} \\
&= \begin{bmatrix} a_{k(0,0)} P_{k-1|k-1(0,2)} & a_{k(0,0)} P_{k-1|k-1(0,3)} \\ a_{k(1,1)} P_{k-1|k-1(1,2)} & a_{k(1,1)} P_{k-1|k-1(1,3)} \end{bmatrix} \\
&\quad + \begin{bmatrix} a_{k(0,2)} P_{k-1|k-1(2,2)} + a_{k(0,3)} P_{k-1|k-1(2,3)} & a_{k(0,2)} P_{k-1|k-1(2,3)} + a_{k(0,3)} P_{k-1|k-1(3,3)} \\ a_{k(1,2)} P_{k-1|k-1(2,2)} + a_{k(1,3)} P_{k-1|k-1(2,3)} & a_{k(1,2)} P_{k-1|k-1(2,3)} + a_{k(1,3)} P_{k-1|k-1(3,3)} \end{bmatrix} \\
(AP_B + BP_D)D^T &= \begin{bmatrix} APpBP_{(0,0)} & APpBP_{(0,1)} \\ APpBP_{(1,0)} & APpBP_{(1,1)} \end{bmatrix} \begin{bmatrix} 1 & a_{k(3,2)} \\ 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} APpBP_{(0,0)} & a_{k(3,2)} APpBP_{(0,0)} + APpBP_{(0,1)} \\ APpBP_{(1,0)} & a_{k(3,2)} APpBP_{(1,0)} + APpBP_{(1,1)} \end{bmatrix} \\
DP_D D^T &= \begin{bmatrix} 1 & 0 \\ a_{k(3,2)} & 1 \end{bmatrix} \begin{bmatrix} P_{k-1|k-1(2,2)} & P_{k-1|k-1(2,3)} \\ P_{k-1|k-1(3,2)} & P_{k-1|k-1(3,3)} \end{bmatrix} \begin{bmatrix} 1 & a_{k(3,2)} \\ 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} P_{k-1|k-1(2,2)} & P_{k-1|k-1(2,3)} \\ a_{k(3,2)} P_{k-1|k-1(2,2)} + P_{k-1|k-1(2,3)} & a_{k(3,2)} P_{k-1|k-1(2,3)} + P_{k-1|k-1(3,3)} \end{bmatrix} \begin{bmatrix} 1 & a_{k(3,2)} \\ 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} P_{k-1|k-1(2,2)} & a_{k(3,2)} P_{k-1|k-1(2,2)} + P_{k-1|k-1(2,3)} \\ a_{k(3,2)} P_{k-1|k-1(2,2)} + P_{k-1|k-1(2,3)} & a_{k(3,2)} a_{k(3,2)} P_{k-1|k-1(2,2)} + 2a_{k(3,2)} P_{k-1|k-1(2,3)} + P_{k-1|k-1(3,3)} \end{bmatrix}
\end{aligned}$$

// 计算 $P_{k_k_prev}$

$$P_{k_k_prev_0_0} = A_{k_0_0} * A_{k_0_0} * P_{k_k_0_0} + 2 * A_{k_0_0} * (A_{k_0_2} * P_{k_k_0_2} + A_{k_0_3} * P_{k_k_0_3})$$

$$+ A_{k_0_2} * A_{k_0_2} * P_{k_k_2_2} + 2 * A_{k_0_2} * A_{k_0_3} * P_{k_k_2_3} + A_{k_0_3} * A_{k_0_3} * P_{k_k_3_3} + Q_{0_0};$$

$$P_{k_k_prev_0_1} = A_{k_0_0} * A_{k_1_1} * P_{k_k_0_1} + A_{k_1_1} * (A_{k_0_2} * P_{k_k_1_2} + A_{k_0_3} * P_{k_k_1_3})$$

$$+ A_{k_0_0} * (A_{k_1_2} * P_{k_k_0_2} + A_{k_1_3} * P_{k_k_0_3})$$

$$+ A_{k_1_2} * (A_{k_0_2} * P_{k_k_2_2} + A_{k_0_3} * P_{k_k_2_3})$$

$$+ A_{k_1_3} * (A_{k_0_2} * P_{k_k_2_3} + A_{k_0_3} * P_{k_k_3_3});$$

$P_{k_k_prev_1_1} = A_{k_1_1} * A_{k_1_1} * P_{k_k_1_1} + 2 * A_{k_1_1} * (A_{k_1_2} * P_{k_k_1_2} + A_{k_1_3} * P_{k_k_1_3})$
 $+ A_{k_1_2} * A_{k_1_2} * P_{k_k_2_2} + 2 * A_{k_1_2} * A_{k_1_3} * P_{k_k_2_3}$
 $+ A_{k_1_3} * A_{k_1_3} * P_{k_k_3_3} + Q_{1_1};$
 $P_{k_k_prev_2_2} = P_{k_k_2_2} + Q_{2_2};$
 $P_{k_k_prev_3_3} = A_{k_3_2} * A_{k_3_2} * P_{k_k_2_2} + 2 * A_{k_3_2} * P_{k_k_2_3} + P_{k_k_3_3} + Q_{3_3};$
 $P_{k_k_prev_2_3} = A_{k_3_2} * P_{k_k_2_2} + P_{k_k_2_3};$

const float

$APpBP_0_0 = A_{k_0_0} * P_{k_k_0_2} + A_{k_0_2} * P_{k_k_2_2} + A_{k_0_3} * P_{k_k_2_3}$
 $, APpBP_0_1 = A_{k_0_0} * P_{k_k_0_3} + A_{k_0_2} * P_{k_k_2_3} + A_{k_0_3} * P_{k_k_3_3}$
 $, APpBP_1_0 = A_{k_1_1} * P_{k_k_1_2} + A_{k_1_2} * P_{k_k_2_2} + A_{k_1_3} * P_{k_k_2_3}$
 $, APpBP_1_1 = A_{k_1_1} * P_{k_k_1_3} + A_{k_1_2} * P_{k_k_2_3} + A_{k_1_3} * P_{k_k_3_3};$
 $P_{k_k_prev_0_2} = APpBP_0_0;$
 $P_{k_k_prev_1_2} = APpBP_1_0;$
 $P_{k_k_prev_0_3} = A_{k_3_2} * APpBP_0_0 + APpBP_0_1;$
 $P_{k_k_prev_1_3} = A_{k_3_2} * APpBP_1_0 + APpBP_1_1;$

卡尔曼增益

$$K_k = P_{k|k-1} H^T (H P_{k|k-1} H^T + R)^{-1} = P_{k|k-1} C^T (C P_{k|k-1} C^T + R)^{-1}, C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$P_{k|k-1} C^T = \begin{bmatrix} P_{k|k-1}(0,0) & P_{k|k-1}(0,1) & P_{k|k-1}(0,2) & P_{k|k-1}(0,3) \\ P_{k|k-1}(0,1) & P_{k|k-1}(1,1) & P_{k|k-1}(1,2) & P_{k|k-1}(1,3) \\ P_{k|k-1}(0,2) & P_{k|k-1}(1,2) & P_{k|k-1}(2,2) & P_{k|k-1}(2,3) \\ P_{k|k-1}(0,3) & P_{k|k-1}(1,3) & P_{k|k-1}(2,3) & P_{k|k-1}(3,3) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} P_{k|k-1}(0,0) & P_{k|k-1}(0,1) \\ P_{k|k-1}(0,1) & P_{k|k-1}(1,1) \\ P_{k|k-1}(0,2) & P_{k|k-1}(1,2) \\ P_{k|k-1}(0,3) & P_{k|k-1}(1,3) \end{bmatrix}$$

$$C P_{k|k-1} C^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_{k|k-1}(0,0) & P_{k|k-1}(0,1) \\ P_{k|k-1}(0,1) & P_{k|k-1}(1,1) \\ P_{k|k-1}(0,2) & P_{k|k-1}(1,2) \\ P_{k|k-1}(0,3) & P_{k|k-1}(1,3) \end{bmatrix} = \begin{bmatrix} P_{k|k-1}(0,0) & P_{k|k-1}(0,1) \\ P_{k|k-1}(0,1) & P_{k|k-1}(1,1) \end{bmatrix}$$

$$(C P_{k|k-1} C^T + R)^{-1} = \begin{bmatrix} P_{k|k-1}(0,0) + R_{(0,0)} & P_{k|k-1}(0,1) \\ P_{k|k-1}(0,1) & P_{k|k-1}(1,1) + R_{(1,1)} \end{bmatrix}^{-1}$$

行列式（这里理论分析其必定可逆，因此行列式不为0）

$$|C P_{k|k-1} C^T + R| = (P_{k|k-1}(0,0) + R_{(0,0)})(P_{k|k-1}(1,1) + R_{(1,1)}) - P_{k|k-1}(0,1)^2$$

$$(C P_{k|k-1} C^T + R)^{-1} = \frac{1}{|C P_{k|k-1} C^T + R|} \begin{bmatrix} P_{k|k-1}(1,1) + R_{(1,1)} & -P_{k|k-1}(0,1) \\ -P_{k|k-1}(0,1) & P_{k|k-1}(0,0) + R_{(0,0)} \end{bmatrix}$$

$$\begin{aligned}
K_k &= P_{k|k-1} C^T (C P_{k|k-1} C^T + R)^{-1} \\
&= \frac{1}{|C P_{k|k-1} C^T + R|} \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} \\ P_{k|k-1(0,1)} & P_{k|k-1(1,1)} \\ P_{k|k-1(0,2)} & P_{k|k-1(1,2)} \\ P_{k|k-1(0,3)} & P_{k|k-1(1,3)} \end{bmatrix} \begin{bmatrix} P_{k|k-1(1,1)} + R_{(1,1)} & -P_{k|k-1(0,1)} \\ -P_{k|k-1(0,1)} & P_{k|k-1(0,0)} + R_{(0,0)} \end{bmatrix} \\
&= \frac{1}{|C P_{k|k-1} C^T + R|} \\
&\quad * \begin{bmatrix} P_{k|k-1(0,0)}(P_{k|k-1(1,1)} + R_{(1,1)}) - P_{k|k-1(0,1)}P_{k|k-1(0,1)} & P_{k|k-1(0,1)}R_{(0,0)} \\ P_{k|k-1(0,1)}R_{(1,1)} & -P_{k|k-1(0,1)}P_{k|k-1(0,1)} + P_{k|k-1(1,1)}(P_{k|k-1(0,0)} + R_{(0,0)}) \\ P_{k|k-1(0,2)}(P_{k|k-1(1,1)} + R_{(1,1)}) - P_{k|k-1(1,2)}P_{k|k-1(0,1)} & -P_{k|k-1(0,1)}P_{k|k-1(0,2)} + P_{k|k-1(1,2)}(P_{k|k-1(0,0)} + R_{(0,0)}) \\ P_{k|k-1(0,3)}(P_{k|k-1(1,1)} + R_{(1,1)}) - P_{k|k-1(1,3)}P_{k|k-1(0,1)} & -P_{k|k-1(0,3)}P_{k|k-1(0,1)} + P_{k|k-1(1,3)}(P_{k|k-1(0,0)} + R_{(0,0)}) \end{bmatrix}
\end{aligned}$$

提取公因子（注意这里为命名一致，PR01 的命名跟随前面 PR00,PR11，有一个 R）：

$$PR_{00} = P_{k|k-1(0,0)} + R_{(0,0)}, PR_{11} = P_{k|k-1(1,1)} + R_{(1,1)}, PR_{01} = P_{k|k-1(0,1)}P_{k|k-1(0,1)}$$

式子可以改为

$$\begin{aligned}
K_k &= \frac{1}{PR_{00}PR_{11} - PR_{01}} \\
&\quad * \begin{bmatrix} P_{k|k-1(0,0)}PR_{11} - PR_{01} & P_{k|k-1(0,1)}R_{(0,0)} \\ P_{k|k-1(0,1)}R_{(1,1)} & -PR_{01} + P_{k|k-1(1,1)}PR_{00} \\ P_{k|k-1(0,2)}PR_{11} - P_{k|k-1(1,2)}P_{k|k-1(0,1)} & -P_{k|k-1(0,1)}P_{k|k-1(0,2)} + P_{k|k-1(1,2)}PR_{00} \\ P_{k|k-1(0,3)}PR_{11} - P_{k|k-1(1,3)}P_{k|k-1(0,1)} & -P_{k|k-1(0,3)}P_{k|k-1(0,1)} + P_{k|k-1(1,3)}PR_{00} \end{bmatrix}
\end{aligned}$$

// 计算 Kk

```

const float PR00=P_k_k_prev_0_0+R_0_0;
const float PR11=P_k_k_prev_1_1+R_1_1;
const float PR01=P_k_k_prev_0_1*P_k_k_prev_0_1;
const float det=1.f/(PR00*PR11-PR01);
K_k[0][0]=det*(P_k_k_prev_0_0*PR11-PR01);
K_k[0][1]=det*(P_k_k_prev_0_1*R_0_0);
K_k[1][0]=det*P_k_k_prev_0_1*R_1_1;
K_k[1][1]=det*(-PR01+P_k_k_prev_1_1*PR00);
K_k[2][0]=det*(P_k_k_prev_0_2*PR11-P_k_k_prev_1_2*P_k_k_prev_0_1);
K_k[2][1]=det*(-P_k_k_prev_0_1*P_k_k_prev_0_2+P_k_k_prev_1_2*PR00);
K_k[3][0]=det*(P_k_k_prev_0_3*PR11-P_k_k_prev_1_3*P_k_k_prev_0_1);
K_k[3][1]=det*(-P_k_k_prev_0_3*P_k_k_prev_0_1+P_k_k_prev_1_3*PR00);

```

后验估计

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - h(\tilde{x}_{k-1}, 0)) = \hat{x}_{k|k-1} + K_k(y_k - C\tilde{x}_{k-1})$$

其中

$$\tilde{x}_{k-1} = f(\hat{x}_{k-1|k-1}, u_k, 0) = \hat{x}_{k|k-1}$$

// ypCx=y-Cx~(k-1)=y-Cf(xk-1 | k-1,uk,0)=y-Cxk | k-1

```

float ypCx[2] = {
    ekf->Ialpha_I - x_k_k_prev[0],
    ekf->Ibeta_I - x_k_k_prev[1]
};

```



```
};
for (uint8_t a = 0; a < 4; ++a)
    x_k_k[a] = x_k_k_prev[a] + K_k[a][0] * ypCx[0] + K_k[a][1] * ypCx[1];
```

更新协方差矩阵

$$P_{k|k} = (I - K_k C) P_{k|k-1}$$

$$K_k C P_{k|k-1} = \begin{bmatrix} K_{k(0,0)} & K_{k(0,1)} & 0 & 0 \\ K_{k(1,0)} & K_{k(1,1)} & 0 & 0 \\ K_{k(2,0)} & K_{k(2,1)} & 0 & 0 \\ K_{k(3,0)} & K_{k(3,1)} & 0 & 0 \end{bmatrix} \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} & P_{k|k-1(0,2)} & P_{k|k-1(0,3)} \\ P_{k|k-1(1,0)} & P_{k|k-1(1,1)} & P_{k|k-1(1,2)} & P_{k|k-1(1,3)} \\ P_{k|k-1(2,0)} & P_{k|k-1(2,1)} & P_{k|k-1(2,2)} & P_{k|k-1(2,3)} \\ P_{k|k-1(3,0)} & P_{k|k-1(3,1)} & P_{k|k-1(3,2)} & P_{k|k-1(3,3)} \end{bmatrix} =$$

$$K_k C P_{k|k-1} = \begin{bmatrix} K_{k(0,0)} & K_{k(0,1)} \\ K_{k(1,0)} & K_{k(1,1)} \\ K_{k(2,0)} & K_{k(2,1)} \\ K_{k(3,0)} & K_{k(3,1)} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} & P_{k|k-1(0,2)} & P_{k|k-1(0,3)} \\ P_{k|k-1(1,0)} & P_{k|k-1(1,1)} & P_{k|k-1(1,2)} & P_{k|k-1(1,3)} \\ P_{k|k-1(2,0)} & P_{k|k-1(2,1)} & P_{k|k-1(2,2)} & P_{k|k-1(2,3)} \\ P_{k|k-1(3,0)} & P_{k|k-1(3,1)} & P_{k|k-1(3,2)} & P_{k|k-1(3,3)} \end{bmatrix}$$

$$= \begin{bmatrix} K_{k(0,0)} & K_{k(0,1)} \\ K_{k(1,0)} & K_{k(1,1)} \\ K_{k(2,0)} & K_{k(2,1)} \\ K_{k(3,0)} & K_{k(3,1)} \end{bmatrix} \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} & P_{k|k-1(0,2)} & P_{k|k-1(0,3)} \\ P_{k|k-1(1,0)} & P_{k|k-1(1,1)} & P_{k|k-1(1,2)} & P_{k|k-1(1,3)} \end{bmatrix}$$

这里无法再继续化简了

$$P_{k|k} = \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} & P_{k|k-1(0,2)} & P_{k|k-1(0,3)} \\ P_{k|k-1(1,0)} & P_{k|k-1(1,1)} & P_{k|k-1(1,2)} & P_{k|k-1(1,3)} \\ P_{k|k-1(2,0)} & P_{k|k-1(2,1)} & P_{k|k-1(2,2)} & P_{k|k-1(2,3)} \\ P_{k|k-1(3,0)} & P_{k|k-1(3,1)} & P_{k|k-1(3,2)} & P_{k|k-1(3,3)} \end{bmatrix}$$

$$- \begin{bmatrix} K_{k(0,0)} & K_{k(0,1)} \\ K_{k(1,0)} & K_{k(1,1)} \\ K_{k(2,0)} & K_{k(2,1)} \\ K_{k(3,0)} & K_{k(3,1)} \end{bmatrix} \begin{bmatrix} P_{k|k-1(0,0)} & P_{k|k-1(0,1)} & P_{k|k-1(0,2)} & P_{k|k-1(0,3)} \\ P_{k|k-1(1,0)} & P_{k|k-1(1,1)} & P_{k|k-1(1,2)} & P_{k|k-1(1,3)} \end{bmatrix}$$

// 更新协方差矩阵

```
P_k_k_0_0=P_k_k_prev_0_0-K_k[0][0]*P_k_k_prev_0_0-K_k[0][1]*P_k_k_prev_0_1;
P_k_k_0_1=P_k_k_prev_0_1-K_k[0][0]*P_k_k_prev_0_1-K_k[0][1]*P_k_k_prev_1_1;
P_k_k_0_2=P_k_k_prev_0_2-K_k[0][0]*P_k_k_prev_0_2-K_k[0][1]*P_k_k_prev_1_2;
P_k_k_0_3=P_k_k_prev_0_3-K_k[0][0]*P_k_k_prev_0_3-K_k[0][1]*P_k_k_prev_1_3;
```

```
P_k_k_1_1=P_k_k_prev_1_1-K_k[1][0]*P_k_k_prev_0_1-K_k[1][1]*P_k_k_prev_1_1;
P_k_k_1_2=P_k_k_prev_1_2-K_k[1][0]*P_k_k_prev_0_2-K_k[1][1]*P_k_k_prev_1_2;
P_k_k_1_3=P_k_k_prev_1_3-K_k[1][0]*P_k_k_prev_0_3-K_k[1][1]*P_k_k_prev_1_3;
```

```
P_k_k_2_2=P_k_k_prev_2_2-K_k[2][0]*P_k_k_prev_0_2-K_k[2][1]*P_k_k_prev_1_2;
P_k_k_2_3=P_k_k_prev_2_3-K_k[2][0]*P_k_k_prev_0_3-K_k[2][1]*P_k_k_prev_1_3;
```

```
P_k_k_3_3=P_k_k_prev_3_3-K_k[3][0]*P_k_k_prev_0_3-K_k[3][1]*P_k_k_prev_1_3;
```

经优化后，一次 EKF 大概需要 17us，可以满足 20khz 的需求